

Match each term to its definition. Terms: **Token** • **Context window** • **Hallucination** • **RAG** • **Temperature** • **Guardrail** • **Quantization** • **Agent**

- A. Rule or mechanism limiting unsafe/invalid model behavior
- B. Randomness control – lower values make output more predictable
- C. Basic unit of text the model reads/writes: a word, subword, or symbol
- D. Plausible but unsupported or false output
- E. Model-driven system that plans, uses tools, and acts
- F. Maximum number of tokens the model can process at once
- G. Retrieving documents before generating, to ground the answer in real sources
- H. Lower-precision weights for faster, cheaper inference

For each task, mark: **(a) well-suited to Claude directly, (b) needs external tools/data, or (c) high hallucination risk – verify independently.**

1. "Summarize this 5-page PDF I uploaded."
2. "What's the current stock price of Apple?"
3. "Write a Python function to reverse a string."
4. "How many employees does [a small local company] have right now?"
5. "Rewrite this paragraph in a more formal tone."
6. "What's $4,827 \times 39,201$?"
7. "Compare these two reviews and say which is more positive."
8. "Give me the exact page number where this fact appears in a book you weren't shown."

1. Ask Claude a specific, fairly obscure or recent factual question (a niche statistic, a small local business, a very recent event) without enabling search.
2. Record the answer.
3. Verify it independently against a source.
4. Re-ask the identical question, this time explicitly telling Claude to search.
5. Write 2–3 sentences comparing the two answers – what changed, and why.